

DETAILED PRODUCTION-READY PROMPT: VECTOR (Unit 5)

Interactive Learning App Generation for TVET DUM10092

EXECUTIVE BRIEF

Topic: VECTOR (Unit 5 - DUM10092 Engineering Mathematics)

Subtopics Covered: 5.1, 5.2, 5.3 (All from syllabus)

Target Level: TVET Diploma (weak math foundation students)

Duration: 45-60 minutes of learning content

Delivery: Single HTML file, responsive, fully interactive

LEARNING OUTCOMES (C1-C3 Competencies)

After completing this app, students should be able to:

C1 - Remember (Identify & Recognize):

- Identify scalar vs vector quantities correctly
- Recognize vector notation (both forms: $x\mathbf{i} + y\mathbf{j}$ and column)
- Distinguish between null, unit, parallel, and equal vectors
- Identify direction and magnitude of given vectors

C2 - Understand (Explain & Express):

- Explain difference between scalar and vector quantities
- Express vectors in component form (i, j notation)
- Explain how to draw directed line segments
- Interpret vector diagrams in Cartesian plane
- Explain vector equality conditions

C3 - Apply (Solve & Construct):

- Apply triangle law to add vectors geometrically
 - Apply parallelogram law to add vectors geometrically
 - Perform scalar multiplication on vectors
 - Add and subtract vectors algebraically
 - Determine if two vectors are parallel or equal
 - Solve real-world vector problems (forces, velocities)
-

DETAILED SYLLABUS MAPPING

UNIT 5: VECTOR

└ 5.1 INTRODUCTION TO VECTOR

└ Subtopic: Scalar vs Vector

- └ Definition of scalar (magnitude only)
- └ Definition of vector (magnitude + direction)
- └ Examples of each
- └ Notation introduction

└ Subtopic: Types of Vectors

- └ Zero/null vector ($\mathbf{0}$ or $\mathbf{0}$)
- └ Unit vector (magnitude = 1)
- └ Parallel vectors (same direction, any magnitude)
- └ Equal vectors (same magnitude & direction)
- └ Negative vector (opposite direction)

└ Subtopic: Vector Notation

- └ Column form: $\begin{bmatrix} x \\ y \end{bmatrix}$
- └ Component form: $x\mathbf{i} + y\mathbf{j}$
- └ Arrow notation: $\overrightarrow{AB}$ or $\mathbf{v}$
- └ Magnitude notation: $|\mathbf{v}|$ or v

└ Subtopic: Scalar Multiplication

- └ Definition: $k \times \mathbf{v}$ (scalar k times vector $\mathbf{v}$)
- └ Effect on magnitude: $|k\mathbf{v}| = |k| \times |\mathbf{v}|$
- └ Effect on direction: positive k (same direction), negative k (opposite)
- └ Examples: $2\mathbf{v}$, $-\mathbf{v}$, $0.5\mathbf{v}$

└ 5.2 VECTOR IN CARTESIAN PLANES

└ Subtopic: Component Form ($\mathbf{i}$, $\mathbf{j}$ notation)

- └ $\mathbf{i}$ -unit vector: direction along x-axis
- └ $\mathbf{j}$ -unit vector: direction along y-axis
- └ Expressing position vectors
- └ From coordinates to component form
- └ From component form to coordinates

└ Subtopic: Drawing Vectors

- └ Direction: angle from positive x-axis
- └ Magnitude: length of arrow
- └ Directed line segment: from point A to point B
- └ Head-and-tail representation

└ Subtopic: Vector Magnitude & Direction

- └ Magnitude formula: $|v| = \sqrt{x^2 + y^2}$
- └ Direction angle: $\theta = \arctan(y/x)$
- └ Real-world applications (speed, velocity)

└ 5.3 ADDITION & SUBTRACTION OF VECTORS

└ Subtopic: Triangle Law

- └ Definition: Place vectors head-to-tail
- └ Result: Vector from start to end
- └ Geometric construction
- └ Algebraic verification

└ Subtopic: Parallelogram Law

- └ Definition: Place vectors tail-to-tail
- └ Complete parallelogram
- └ Diagonal is resultant
- └ Equivalence to triangle law

└ Subtopic: Algebraic Addition

- └ Add components separately: $(a+c)i + (b+d)j$
- └ Column form addition: $\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a+c \\ b+d \end{bmatrix}$
- └ Properties: commutative, associative
- └ Verification with geometric methods

└ Subtopic: Subtraction & Vector Properties

- └ Subtraction as addition of negative: $A - B = A + (-B)$
- └ Reverse direction of second vector
- └ Parallel vectors (cannot be "split" further)
- └ Non-parallel vectors (unique resultant)
- └ Real-world: relative velocity, net force

CVSA PROGRESSION FOR VECTOR

CONCRETE STAGE (Why vectors matter - real stories)

Story 1: Airplane Flight Path

Scenario: Airplane wants to fly directly North at 300 km/h

Reality: Wind pushes from West at 50 km/h

Question: What's the actual path the airplane takes?

Why it matters:

- Can't just add $300 + 50 = 350$ km/h
- Direction of wind matters (perpendicular to intended path)
- Magnitude AND direction critical
- Pilot must know actual path to reach destination

Visual: Animated airplane with:

- Intended velocity vector (North, 300 km/h) - RED
- Wind velocity vector (East, 50 km/h) - BLUE
- Actual path (resultant, Northeast angle) - GREEN
- Show "if we add like scalars: wrong path"
- Show "if we add like vectors: correct path"

Story 2: Building Forces

Scenario: Construction hook has 2 cables pulling at angles

Cable 1: 800 N force at 30° from horizontal

Cable 2: 600 N force at 120° from horizontal

Question: What's the total force on the hook?

Why it matters:

- Engineers MUST calculate resultant force
- Safety depends on correct force calculation
- Wrong calculation $\rightarrow$ structural failure
- Shows why vector addition (not scalar) is needed

Visual: 2D top-down view of hook with:

- Cable 1 force (RED arrow)
- Cable 2 force (BLUE arrow)
- Resultant force (GREEN arrow showing net pull direction)

Interactive Element (Concrete):

Student drags arrows on canvas to represent vectors:

- Drag arrow A (RED) to length 300, direction North
- Drag arrow B (BLUE) to length 50, direction East
- See resultant (GREEN) automatically drawn
- Understand: changing magnitude/direction changes result

Key insight: "You can't just ADD the numbers!"

VISUAL STAGE (Diagrams showing structure)

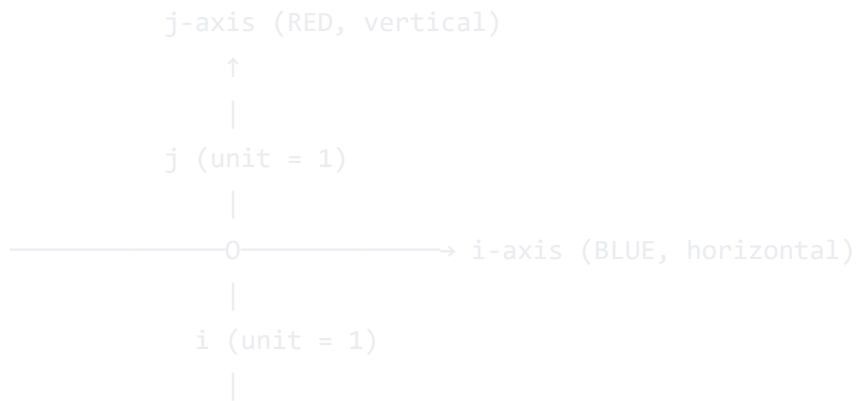
Visual 1: Scalar vs Vector Comparison

SCALAR (Temperature)	VECTOR (Velocity)
25°C (just a number)	50 km/h North (magnitude + direction)
No direction needed No spatial component Cannot be broken into parts	N ↑ ← E W → Arrow shows direction & size

Color-coding throughout app:

- Scalar values: BLACK text (no color)
- Vectors in i-direction: BLUE (always blue for i)
- Vectors in j-direction: RED (always red for j)
- Resultant vectors: GREEN

Visual 2: Cartesian Plane with Unit Vectors



Example: Vector $v = 3i + 2j$

└ Component in i-direction: 3 units BLUE (horizontal)

└ Component in j-direction: 2 units RED (vertical)

Position: Ends at point (3, 2) on Cartesian plane

Visual 3: Vector Equality & Properties

EQUAL VECTORS (same magnitude & direction):

$AB = CD = EF$

[All same length, same direction - shown as parallel arrows]

PARALLEL VECTORS (same direction, different magnitudes):

v and $2v$ are parallel ($2v$ is longer)

NULL VECTOR:

$\vec{0}$ or $\vec{0}$ (point, no length, no direction)

$\vec{0}$

UNIT VECTOR:

$|u| = 1$ (magnitude exactly 1)

Visual 4: Triangle Law Animation

Step 1: Draw vector A from O to P



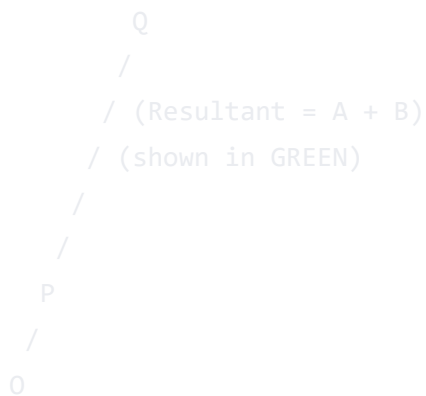
(A drawn, shown in RED)

Step 2: From P, draw vector B to Q



(B's tail at A's head)

Step 3: Resultant from O to Q



(Closes the triangle)

Key: "Head-to-tail = Triangle Law"

Visual 5: Parallelogram Law Animation

Step 1: Draw A from O to A'

Step 2: Draw B from O to B'

Step 3: Complete parallelogram

Step 4: Diagonal OC is resultant (A + B)



Diagonal = same as triangle law result (GREEN)

Interactive Element (Visual):

Student adjusts Cartesian plane:

- Drag point P around
- See i-component (blue line) change length
- See j-component (red line) change length
- Watch vector OV update in real-time
- Understand component breakdown visually

SYMBOL STAGE (Notation and formulas)

Notation Introduction:

VECTOR NOTATION FORMS:

Form 1: Position Vector (from origin O to point P)

OP or $\mathbf{v}$

Form 2: Component Form (i, j basis)

$\mathbf{v} = 3\mathbf{i} + 2\mathbf{j}$ (read: "3 i plus 2 j")

Form 3: Column Vector

$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Form 4: Magnitude (length of vector)

$|\mathbf{v}| = \sqrt{3^2 + 2^2} = \sqrt{13}$

Form 5: Direction (angle from positive x-axis)

$\theta = \arctan(2/3) = 33.7^\circ$

All forms describe the SAME vector:

"3 units East AND 2 units North"

Key Formulas:

1. COMPONENT FORM to COORDINATES:

$$v = 3i + 2j \rightarrow \text{Point is at } (3, 2)$$

2. COORDINATES to COMPONENT FORM:

$$\text{Point } (5, -1) \rightarrow v = 5i - 1j$$

3. MAGNITUDE:

$$|v| = \sqrt{x^2 + y^2}$$

$$\text{Example: } v = 3i + 4j \rightarrow |v| = \sqrt{9 + 16} = \sqrt{25} = 5$$

4. UNIT VECTOR (direction only):

$$u = v / |v|$$

$$\text{Example: } v = 3i + 4j, |v| = 5 \rightarrow u = (3/5)i + (4/5)j$$

5. SCALAR MULTIPLICATION:

$$kv = k(xi + yj) = (kx)i + (ky)j$$

$$\text{Example: } 2(3i + 4j) = 6i + 8j$$

6. VECTOR ADDITION:

$$A + B = (a_1i + a_2j) + (b_1i + b_2j) = (a_1+b_1)i + (a_2+b_2)j$$

$$\text{Example: } (3i + 4j) + (2i - 1j) = 5i + 3j$$

7. VECTOR SUBTRACTION:

$$A - B = (a_1i + a_2j) - (b_1i + b_2j) = (a_1-b_1)i + (a_2-b_2)j$$

$$\text{Example: } (5i + 3j) - (2i - 1j) = 3i + 4j$$

8. PARALLEL VECTORS:

$$A \parallel B \text{ if } A = kB \text{ for some scalar } k$$

$$\text{Example: } (6i + 4j) \parallel (3i + 2j) \text{ because } (6i + 4j) = 2(3i + 2j)$$

9. EQUAL VECTORS:

$$A = B \text{ if they have same } i\text{-component AND same } j\text{-component}$$

$$A = (3i + 4j) \text{ and } B = (3i + 4j) \rightarrow A = B$$

Interactive Element (Symbol):

Formula Input Practice:

Student sees: "Express this vector in i,j form"

[Cartesian diagram showing vector to point (3, 4)]

Student types: $\begin{bmatrix} 3 \\ \end{bmatrix} i + \begin{bmatrix} 4 \\ \end{bmatrix} j$
(input) (input)

Live validation:

- As student types "3" → first box turns YELLOW
- Correct → box turns GREEN, shows ✓
- Wrong → box turns RED, shows X

ABSTRACT STAGE (Independent problem solving)

Problem Type 1: Express Vector in Component Form

Difficulty EASY:

Given: Point P at (2, 5)

Task: Express OP in component form

Answer: $2i + 5j$

Difficulty MEDIUM:

Given: Vector from A(2, 3) to B(7, 8)

Task: Express AB in component form

Working: $AB = (7-2)i + (8-3)j = 5i + 5j$

Answer: $5i + 5j$

Difficulty HARD:

Given: Point P at (3.5, -2.8), unit vectors i and j given

Task: Express OP and find magnitude

Working: $OP = 3.5i - 2.8j$

$$|OP| = \sqrt{(3.5)^2 + (-2.8)^2} = \sqrt{12.25 + 7.84} = \sqrt{20.09} \approx 4.48$$

Answer: Vector = $3.5i - 2.8j$, Magnitude = 4.48 units

Problem Type 2: Add Vectors (Algebraically)

Difficulty EASY:

$$A = 2i + 3j$$

$$B = 1i + 2j$$

Find: $A + B$

Step-by-step:

$$\text{i-components: } 2 + 1 = 3$$

$$\text{j-components: } 3 + 2 = 5$$

$$\text{Result: } 3i + 5j$$

Difficulty MEDIUM:

$$A = 5i + 2j$$

$$B = -3i + 4j$$

Find: $A + B$

Step-by-step:

$$\text{i-components: } 5 + (-3) = 2$$

$$\text{j-components: } 2 + 4 = 6$$

$$\text{Result: } 2i + 6j$$

Difficulty HARD:

$$A = 7i - 5j$$

$$B = -2i + 8j$$

$$C = 3i - 2j$$

Find: $A + B - C$

Step-by-step:

$$\text{i-components: } 7 + (-2) - 3 = 2$$

$$\text{j-components: } -5 + 8 - (-2) = 5$$

$$\text{Result: } 2i + 5j$$

Problem Type 3: Vector Equality Check

Difficulty EASY:

Given: $X = 3i + 4j$, $Y = 3i + 4j$

Question: Are X and Y equal?

Answer: Yes (i-components match, j-components match)

Difficulty MEDIUM:

Given: $A = 2i - 5j$, $B = 2i - 5j$

Question: Are A and B equal? Explain.

Answer: Yes, identical in magnitude and direction

Difficulty HARD:

Given: $P = 4i + 3j$, $Q = 6i + 4.5j$

Question: Are P and Q parallel? Equal? Explain.

Analysis: $6i + 4.5j = 1.5(4i + 3j) = 1.5P$

Answer: Parallel ($Q = 1.5P$) but NOT equal (different magnitudes)

Problem Type 4: Scalar Multiplication

Difficulty EASY:

$$v = 2i + 3j$$

Find: $2v$ and $-v$

Solution:

$$2v = 2(2i + 3j) = 4i + 6j$$

$$-v = -1(2i + 3j) = -2i - 3j$$

Difficulty MEDIUM:

$$u = 3i - 2j$$

Find: $-3u$ and $0.5u$

Solution:

$$-3u = -3(3i - 2j) = -9i + 6j$$

$$0.5u = 0.5(3i - 2j) = 1.5i - 1j$$

Difficulty HARD:

$$v = 3i + 4j$$

Find: k such that $|kv| = 10$

Solution:

$$|kv| = |k| \times |v|$$

$$|k| \times \sqrt{9+16} = 10$$

$$|k| \times 5 = 10$$

$$|k| = 2$$

$$\text{So } k = 2 \text{ or } k = -2$$

Problem Type 5: Real-World Application

Difficulty EASY:

A boat travels 4 km East then 3 km North.

Express total displacement as vector.

Answer: $4\mathbf{i} + 3\mathbf{j}$ km

Difficulty MEDIUM:

A plane flies at 300 km/h North.

Wind blows at 50 km/h East.

Find resultant velocity vector.

Solution:

Plane velocity: $0\mathbf{i} + 300\mathbf{j}$

Wind velocity: $50\mathbf{i} + 0\mathbf{j}$

Resultant: $50\mathbf{i} + 300\mathbf{j}$ km/h

Difficulty HARD:

Two forces act on object:

$F_1 = 12\mathbf{i} + 5\mathbf{j}$ N

$F_2 = -8\mathbf{i} + 3\mathbf{j}$ N

Find resultant force and its magnitude.

Solution:

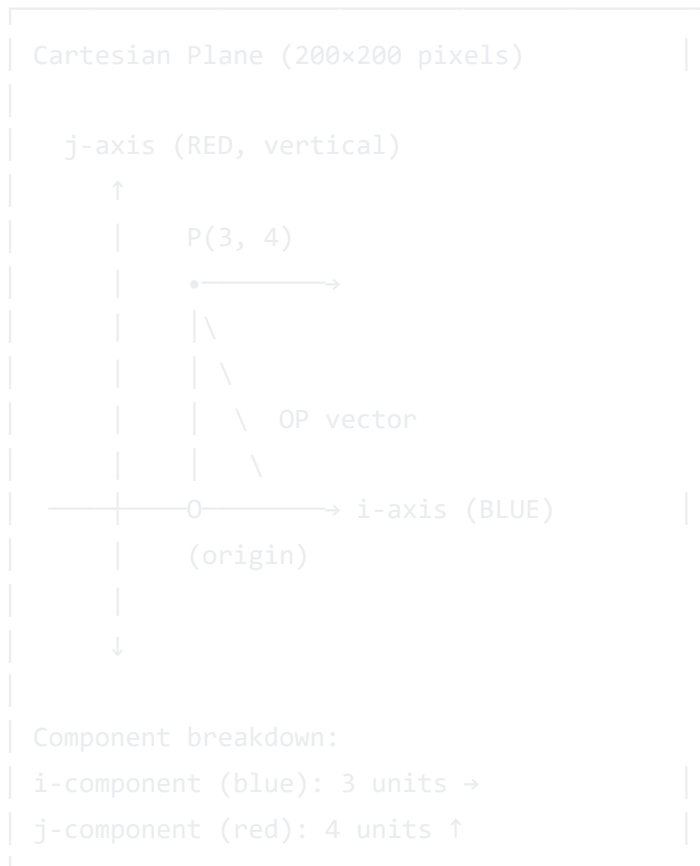
Resultant = $F_1 + F_2 = (12-8)\mathbf{i} + (5+3)\mathbf{j} = 4\mathbf{i} + 8\mathbf{j}$ N

Magnitude = $\sqrt{(16 + 64)} = \sqrt{80} = 4\sqrt{5} \approx 8.94$ N

🎮 INTERACTIVE PATTERNS (Framework 2 Specifications)

PATTERN 1: Express Vector in Component Form

VISUAL DISPLAY:



STUDENT INPUT:

Express vector OP in component form:

$$OP = \begin{bmatrix} 3 \end{bmatrix} i + \begin{bmatrix} 4 \end{bmatrix} j$$

(input) (input)

[Check Answer] [Hint] [Show Solution]

REAL-TIME VALIDATION:

- As student types "3" → first box: YELLOW outline
- Student correct → GREEN outline + ✓
- Student wrong → RED outline + ✗

FEEDBACK (Correct):

✓ Correct! Vector $OP = 3i + 4j$
Meaning: 3 units East (i-direction) + 4 units North (j-direction)"

FEEDBACK (Incorrect):

"X Not quite. Let me help:

- Count blue line (i-component): How many units?
- Count red line (j-component): How many units?

Hint: Looking at the diagram, point P is at $x = \underline{\hspace{1cm}}$, $y = \underline{\hspace{1cm}}$ "

DIFFICULTY PROGRESSION:

Easy: P at (2, 5), positive integers

Medium: P at (-3, 4), with negative

Hard: P at (3.5, -2.7), with decimals

HINTS (4-level):

Hint 1: "Look at the BLUE line (i-direction). How many units?"

Hint 2: "The point is 3 units to the right, so i-component is 3"

Hint 3: "Now look at RED line (j-direction). How many units up?"

Show Answer: "Vector $OP = 3i + 4j$ (3 units right, 4 units up)"

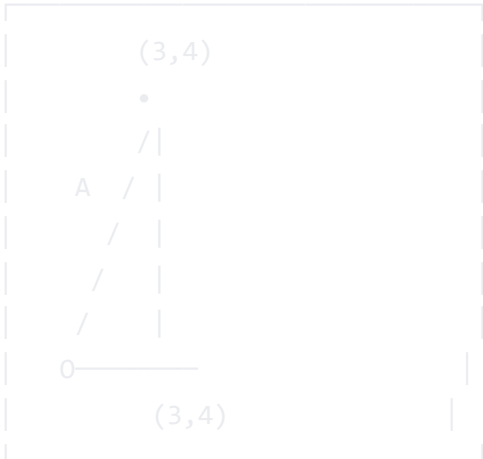
PATTERN 2: Vector Addition Step-by-Step

VISUAL DISPLAY:

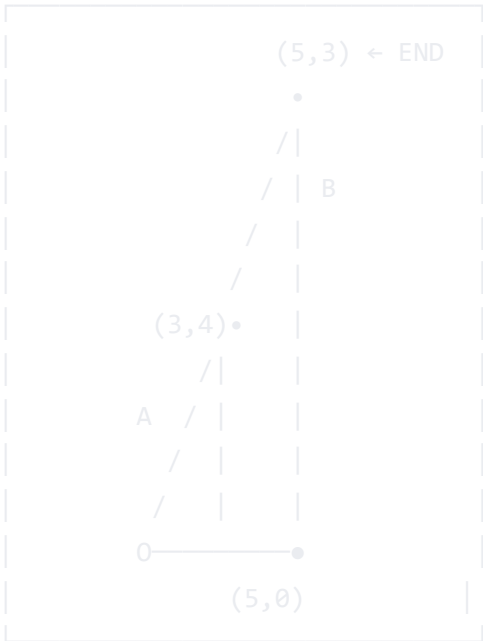
VECTOR ADDITION using TRIANGLE LAW

Given: $A = 3i + 4j$ (RED)
 $B = 2i - 1j$ (BLUE)

Step 1: Draw A from 0 to point (3, 4)

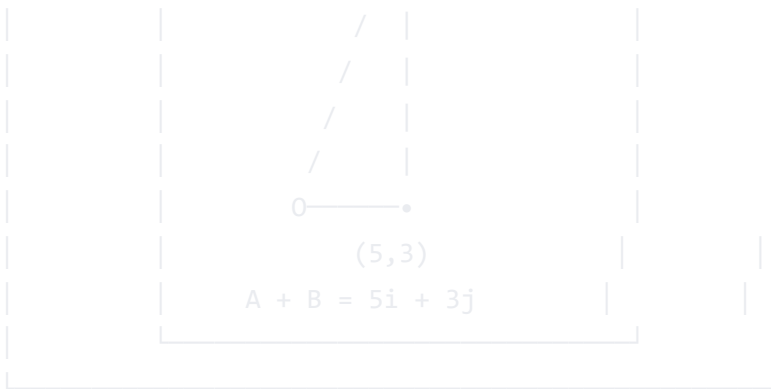


Step 2: From end of A, draw B



Step 3: Resultant from 0 to final point





STUDENT INTERACTIVE TASKS:

STEP 1: Add i-components
 $3 + 2 = [\quad 5 \quad]$

STEP 2: Add j-components
 $4 + (-1) = [\quad 3 \quad]$

STEP 3: Write resultant
 $A + B = [\quad 5 \quad] i + [\quad 3 \quad] j$

VALIDATION LOGIC:

For each step:

- Check answer immediately upon input
- Green if correct, red if incorrect
- Partial credit: "i-component correct, check j-component"
- Cannot move to next step until current step correct

FEEDBACK (Each Step):

Step 1 Correct: "✓ Yes! $3 + 2 = 5$ "

Step 2 Correct: "✓ Good! $4 + (-1) = 3$ (remember: positive plus negative)"

Step 3 Correct: "✓ Perfect! $A + B = 5i + 3j$ "

HINTS (Progressive):

For Step 2:

Hint 1: "Add the numbers: $4 + (-1) = ?$ "

Hint 2: "Think: $4 - 1 = 3$ "

Hint 3: "Positive and negative: $4 + (-1) = 3$ "

SOLUTION REVEAL (Full Jalan Kerja):

"Given: $A = 3i + 4j$ and $B = 2i - 1j$

Step 1: Add i-components

i-component of A: 3

i-component of B: 2

Sum: $3 + 2 = 5$

Step 2: Add j-components

j-component of A: 4

j-component of B: -1

Sum: $4 + (-1) = 3$

Step 3: Write resultant vector

$A + B = 5i + 3j$

Verification (Geometric):

Using triangle law:

- Start at 0, move 3 units East, 4 units North → reach (3,4)
- From (3,4), move 2 units East, 1 unit South → reach (5,3)
- Direct path 0 to (5,3) = 5 units East, 3 units North
- Therefore $A + B = 5i + 3j$ ✓"

DIFFICULTY LEVELS:

Easy: $A = 2i + 1j$, $B = 1i + 2j$ (positive integers)

Medium: $A = 5i - 2j$, $B = -3i + 4j$ (negative included)

Hard: $A = 7.5i - 3.2j$, $B = -2.1i + 5.8j$ (decimals)

PATTERN 3: Vector Equality Determination

VISUAL COMPARISON:

Are these vectors equal?

Vector X = 4i - 2j

Vector Y = 4i - 2j

j

↑

•

/|

/|

/|

0→ i

(4,-2)

Diagram 1

j

↑

•

/|

/|

/|

0→ i

(4,-2)

Diagram 2

COMPARISON TABLE (Auto-updating as student checks):

Component	X	Y	Match? (✓/X)
i-component	4	4	✓ YES
j-component	-2	-2	✓ YES
VERDICT			X = Y (EQUAL)

STUDENT INTERACTION:

Are X and Y equal?

[EQUAL]

[NOT EQUAL]

← Toggle buttons

Your answer: _____

[Submit Answer]

[Get Hint]

[Show Solution]

FEEDBACK (Correct):

"✓ Correct! X and Y are EQUAL.

Both have i-component = 4 AND j-component = -2

Equal vectors have same magnitude AND same direction."

FEEDBACK (Incorrect):

"X Not quite. Let's check:

X has i = 4, j = -2

Y has i = 4, j = -2

All components match, so they ARE equal."

HINTS:

Hint 1: "Compare each component. Do all match?"

Hint 2: "Check i-components: X has 4, Y has 4. Match? Yes"

Hint 3: "Check j-components: X has -2, Y has -2. Match? Yes"

Show Answer: "Both vectors are EQUAL (same i and j components)"

DIFFICULTY LEVELS:

Easy: $X = 3i + 4j$, $Y = 3i + 4j$ (obviously equal)

Medium: $X = 5i - 2j$, $Y = 5i + 2j$ (one component different)

Hard: $X = 6i + 8j$, $Y = 3i + 4j$ (parallel but not equal)

PATTERN 4: Dynamic Problem Generator

JAVASCRIPT PSEUDOCODE:

```
function generateVectorProblem(difficulty, problemType) {

  if (difficulty === 'easy') {
    // Small positive integers (1-5)
    const x1 = Math.floor(Math.random() * 5) + 1;
    const y1 = Math.floor(Math.random() * 5) + 1;

    if (problemType === 'express') {
      return {
        type: 'express_vector',
        point: { x: x1, y: y1 },
        instruction: `Express vector O→P in i,j form where P(${x1}, ${y1})`,
        correctAnswer: { i: x1, j: y1 }
      };
    } else if (problemType === 'add') {
      const x2 = Math.floor(Math.random() * 5) + 1;
      const y2 = Math.floor(Math.random() * 5) + 1;

      return {
        type: 'add_vectors',
        vectorA: { i: x1, j: y1 },
        vectorB: { i: x2, j: y2 },
        instruction: `Add: (${x1}i + ${y1}j) + (${x2}i + ${y2}j)`,
        correctAnswer: { i: x1 + x2, j: y1 + y2 }
      };
    }
  }

  else if (difficulty === 'medium') {
    // Mix of positive and negative (-10 to 10)
    const x1 = Math.floor(Math.random() * 21) - 10; // -10 to 10
    const y1 = Math.floor(Math.random() * 21) - 10;
    const x2 = Math.floor(Math.random() * 21) - 10;
    const y2 = Math.floor(Math.random() * 21) - 10;

    const operation = Math.random() > 0.5 ? 'add' : 'subtract';

    if (operation === 'add') {
      return {
        type: 'add_vectors',
        vectorA: { i: x1, j: y1 },
```

```

        vectorB: { i: x2, j: y2 },
        instruction: `Add: (${x1}i ${y1}>=0?'+'':''} ${y1}j) + (${x2}i
${y2}>=0?'+'':''} ${y2}j)`,
        correctAnswer: { i: x1 + x2, j: y1 + y2 }
    };
} else {
    return {
        type: 'subtract_vectors',
        vectorA: { i: x1, j: y1 },
        vectorB: { i: x2, j: y2 },
        instruction: `Subtract: (${x1}i ${y1}>=0?'+'':''} ${y1}j) - (${x2}i
${y2}>=0?'+'':''} ${y2}j)`,
        correctAnswer: { i: x1 - x2, j: y1 - y2 }
    };
}
}

else if (difficulty === 'hard') {
    // Decimals, real-world context
    const x1 = (Math.random() * 20 - 10).toFixed(1); // -10 to 10
    const y1 = (Math.random() * 20 - 10).toFixed(1);

    const problemTypes = [
        {
            type: 'resultant_force',
            scenario: 'Two forces acting on object',
            F1_i: parseFloat(x1),
            F1_j: parseFloat(y1),
            F2_i: (Math.random() * 20 - 10).toFixed(1),
            F2_j: (Math.random() * 20 - 10).toFixed(1),
            unit: 'N'
        },
        {
            type: 'resultant_velocity',
            scenario: 'Boat crossing river with current',
            boatI: parseFloat(x1),
            boatJ: parseFloat(y1),
            currentI: (Math.random() * 10 - 5).toFixed(1),
            currentJ: 0,
            unit: 'km/h'
        }
    ];

    return problemTypes[Math.floor(Math.random() * problemTypes.length)];
}

```

```
}

// Usage:
let problem = generateVectorProblem('medium', 'add');
displayProblem(problem);

// When student submits:
let userAnswer = getUserInput(); // { i: ___, j: ___ }
let isCorrect = checkAnswer(userAnswer, problem.correctAnswer);
```

APP STRUCTURE (6 Tabs)

TAB 1: INTRO - "Why Vectors Matter?"

Section 1: Real-World Story 1 - Airplane Flight

HEADING: "Pesawat Terbang Melintasi Angin (Airplane Flying Through Wind)"

SCENARIO:

"Pilot ingin terbang langsung ke Utara dengan kecepatan 300 km/j.
Tetapi angin berhembus dari Barat dengan kecepatan 50 km/j.
Soalan: Kemana sebenarnya pesawat akan terbang?"

VISUAL: Canvas animation showing:

- Velocity vector (North, 300 km/h) - RED arrow
- Wind vector (East, 50 km/h) - BLUE arrow
- Resultant path - GREEN arrow at angle

INTERACTIVE:

Student can:

- Drag wind speed slider (0-100 km/h)
- See airplane path change in real-time
- Understand: "Arah angin mempengaruhi laluan sebenar"

KEY INSIGHT:

"Anda tidak boleh hanya menambah $300 + 50 = 350$ km/h.
Arah sangat penting!
Inilah kenapa kita pelajari VEKTOR (bukan SKALAR)."

Section 2: Real-World Story 2 - Building Forces

HEADING: "Kabel Konstruksi (Construction Cable Forces)"

SCENARIO:

"Dua kabel menarik hook di lokasi pembinaan.

Kabel 1: 800 N pada sudut 30° dari horizontal

Kabel 2: 600 N pada sudut 120° dari horizontal

Soalan: Apakah daya paduan pada hook?"

VISUAL: Top-down diagram showing:

- Hook in center
- Two cables pulling at different angles
- Resultant force shown dynamically

INTERACTIVE:

Student adjusts:

- Magnitude of Force 1 (slider)
- Angle of Force 1 (slider)
- See resultant change visually

KEY INSIGHT:

"Jurutera MESTI mengira daya paduan dengan betul.

Jika salah, struktur boleh runtuh!

Itulah kuasa VEKTOR dalam dunia nyata."

[Button] Mulai Pembelajaran (Start Learning)

TAB 2:  CONCEPT 1 - "Skalar vs Vektor (Scalar vs Vector)"

Definition Section:

HEADING: "Apakah Perbezaan Skalar dan Vektor?"

SKALAR (Scalar):

Definition: "Kuantiti yang hanya mempunyai magnitud (saiz/nilai)"

Examples:

- Suhu: 25°C (hanya angka, tiada arah)
- Jarak: 50 m (berapa jauh, tapi tidak ke mana)
- Masa: 2 jam (berapa lama saja)
- Jisim: 70 kg (berapa berat saja)

VEKTOR (Vector):

Definition: "Kuantiti yang mempunyai magnitud DAN arah"

Examples:

- Halaju: 50 km/j ke Utara (berapa cepat DAN ke mana)
- Sesaran: 100 m ke Timur (berapa jauh DAN arah mana)
- Daya: 500 N ke bawah (kuatnya berapa DAN arah mana)

COMPARISON TABLE:

Ciri-ciri	SKALAR	VEKTOR
Magnitud	Ya (saja)	Ya + Arah
Arah	Tidak	Ya (penting)
Contoh notasi	5 m/s	5 m/s ke Utara
Penambahan	$3 + 2 = 5$	Mesti guna gambar
Boleh diuraikan	Tidak	Ya (i, j components)

Interactive Quiz:

TASK: Kategori setiap kuantiti sebagai SKALAR atau VEKTOR

Items to categorize:

- Suhu bilik 20°C → [SCALAR] [VECTOR]
- Kecepatan 30 km/j timur → [SCALAR] [VECTOR]
- Jarak 5 km → [SCALAR] [VECTOR]
- Sesaran 5 km utara → [SCALAR] [VECTOR]
- Daya 200 N → [SCALAR] [VECTOR]
- Daya 200 N ke bawah → [SCALAR] [VECTOR]

Real-time feedback:

✓ Correct: "Betul! Suhu hanya angka, tiada arah (Skalar)"

✗ Wrong: "Tidak. Perhatian: Arah disebutkan (Vektor)"

TAB 3: CONCEPT 2 - "Vektor dalam Satah Cartes (Vector in Cartesian Plane)"

Component Form Introduction:

HEADING: "Bentuk Komponen: $x_i + y_j$ "

EXPLANATION:

i-unit vector: Menunjuk ke arah positif x (TIMUR)
j-unit vector: Menunjuk ke arah positif y (UTARA)

Vektor $v = 3i + 2j$ bermaksud:
"3 unit ke TIMUR (i-direction) + 2 unit ke UTARA (j-direction)"

VISUAL: Cartesian plane dengan color-coding:

- Horizontal line (3 units): BLUE (i-direction)
- Vertical line (2 units): RED (j-direction)
- Arrow from 0 to endpoint: GREEN (resultant)

POINT COORDINATES ↔ VECTOR FORM:

Jika titik P pada (3, 2):
→ Vektor $OP = 3i + 2j$

Jika vektor $v = 5i - 2j$:
→ Titik akhir pada (5, -2)

Interactive Converter:

VISUAL: Cartesian plane dengan:

- Horizontal grid (0-10)
- Vertical grid (-5 to 5)
- Draggable point P
- Live display of coordinates and vector form

Student drags point P:

- When P moves → coordinates update
- When coordinates update → vector form updates
- Shows i-component (blue) and j-component (red)

Example transitions:

Coordinates (2, 3) ↔ Vector $2\mathbf{i} + 3\mathbf{j}$

Coordinates (5, -1) ↔ Vector $5\mathbf{i} - 1\mathbf{j}$

Coordinates (-3, 4) ↔ Vector $-3\mathbf{i} + 4\mathbf{j}$

Understanding: Same vector can be translated (moved) anywhere, but keeps same i and j components.

Practice: Express Vectors

TASK 1: Given coordinates, write vector

Point: (3, 4)

Answer: $[3]\mathbf{i} + [4]\mathbf{j}$

(with real-time validation)

TASK 2: Given vector, identify point

Vector: $2\mathbf{i} - 5\mathbf{j}$

Answer: Point is at $([2], [-5])$

(with real-time validation)

TASK 3: Calculate magnitude

Vector: $3\mathbf{i} + 4\mathbf{j}$

Task: Find $|\mathbf{v}| = \sqrt{3^2 + 4^2} = \sqrt{[___]} = [___]$

Steps: $3^2 = [9]$, $4^2 = [16]$, $9+16 = [25]$, $\sqrt{25} = [5]$

TAB 4: CONCEPT 3 - "Penambahan & Pengurangan Vektor (Addition & Subtraction)"

Triangle Law Visual:

HEADING: "Hukum Segitiga (Triangle Law)"

DEFINITION:

"Untuk menambah vektor A dan B:

1. Lukiskan A dari titik O
2. Dari hujung A, lukiskan B
3. Vektor paduan (A+B) dari O ke hujung B"

VISUAL ANIMATION (3 steps):

Step 1: A appears (red arrow from O)

Step 2: B appears (blue arrow from A's end)

Step 3: Resultant highlighted (green arrow O to B's end)

Key insight: "Bentuk SEGI TIGA"

Parallelogram Law Visual:

HEADING: "Hukum Segiempat Selari (Parallelogram Law)"

DEFINITION:

"Untuk menambah vektor A dan B:

1. Lukiskan A dari titik O
2. Dari O, lukiskan B juga
3. Lengkapkan segi empat selari
4. Vektor paduan adalah pepenjuru"

VISUAL ANIMATION (4 steps):

Step 1: A from O (red)

Step 2: B from O (blue)

Step 3: Parallel lines drawn

Step 4: Diagonal highlighted (green resultant)

Comparison: "Triangle Law and Parallelogram Law give SAME result!"

Interactive Problem Solver:

HEADING: "Kira Hasil Tambah (Calculate Resultant)"

Given: $A = 3i + 4j$, $B = 2i - 1j$

Find: $A + B$

STEP-BY-STEP GUIDE:

LANGKAH 1: Tambah komponen i

Komponen i bagi A: 3

Komponen i bagi B: 2

Hasil: $3 + 2 = [5]$ ← Student inputs

[Check] [Hint] [Show]

LANGKAH 2: Tambah komponen j

Komponen j bagi A: 4

Komponen j bagi B: -1

Hasil: $4 + (-1) = [3]$ ← Student inputs

[Check] [Hint] [Show]

LANGKAH 3: Vektor Paduan

$A + B = [5]i + [3]j$ ← Student inputs both

[Check Answer] [Show Full Solution]

VISUALISASI:

- Diagram showing triangle law with A (red), B (blue), A+B (green)
- All components labeled with colors
- Head-to-tail connection clear

TAB5: 🎮 PRACTICE - "Latihan Interaktif (Interactive Practice)"

Difficulty Selector:

BUTTONS: [Mudah / Easy] [Sederhana / Medium] [Mencabar / Hard]

Easy Tab Selected:

"Soalan 1 of 5 | Tepat: 0 | Peratus: 0%"

PROBLEM GENERATION:

Each problem randomly generated from:

- Type: Express vector / Add vectors / Vector equality / Scalar multiplication
- Difficulty level: Positive integers / With negatives / With decimals

Real-time score tracking visible

Problem Display:

HEADING: "Soalan 1 (Question 1)"

DIAGRAM: Cartesian plane or problem setup

TEXT: Problem statement (Bahasa Malaysia)

INPUT FIELDS:
[Answer box 1] [Answer box 2] (as needed)

BUTTONS:
[Semak Jawapan (Check Answer)]
[Petua (Hint)]
[Tunjukkan Penyelesaian (Show Solution)]
[Soalan Seterusnya (Next Problem)]

INSTANT FEEDBACK:
Green box + ✓ → Correct!
Red box + ✗ → Try again

PROGRESS TRACKER:
Score: 2/5
Accuracy: 40%
XP: 45/100 to reach Level 2
[Progress bar]

GAMIFICATION:
Each correct answer:
- First try: +10 XP, "Perfect! Master mode!"
- Second try: +7 XP, "Good effort!"
- After hint: +5 XP, "Nice work!"

TAB 6: 💡 TIPS - "Kesilapan Lazim (Common Mistakes)"

Mistake 1:

✗ WRONG:

"Komponen i dan j boleh ditambah bersama"

$3i + 4j = 7$ (treating i and j like regular variables)

✓ CORRECT:

"Komponen i dan j BERBEZA - hanya boleh tambah sesama jenis"

$(3i + 2i) + (4j + 5j) = 5i + 9j$

TIDAK boleh: $3i + 4j = 7$ (ini salah!)

Analogi:

"3 buku + 4 pensil $\neq$ 7. Jenis berbeza!"

Begitu juga: $3i + 4j$ tidak boleh dijumlahkan seperti angka biasa"

Mistake 2:

✗ WRONG:

"Untuk tambah vektor, cukup tambah magnitud"

$|A| = 5, |B| = 3 \rightarrow |A+B| = 8$

✓ CORRECT:

"Kena guna hukum segitiga atau segiempat selari"

Vektor A = $3i + 4j$ dengan $|A| = 5$

Vektor B = $4i + 0j$ dengan $|B| = 4$

$A + B = 7i + 4j$ dengan $|A+B| = \sqrt{49+16} = \sqrt{65} \approx 8.06$

(Kebetulan hampir sama, tapi CARA BERBEZA!)

Sebab: Arah vektor penting, bukan hanya magnitud"

Mistake 3:

✗ WRONG:

"Komponen negatif bermakna tidak ada"

$-3i + 4j$: "Arah i tidak ada"

✓ CORRECT:

"Negatif bermakna arah SEBALIKNYA"

$-3i$ means: 3 units ke BARAT (opposite of TIMUR)

$+4j$ means: 4 units ke UTARA

Jadi vektor: 3 units BARAT + 4 units UTARA

Analogi:

" $+3$ pada i-axis $\rightarrow$ 3 steps TIMUR

-3 pada i-axis $\rightarrow$ 3 steps BARAT (sebaliknya)"

Quick Reference Formulas:

1. COMPONENT FORM:	Point (x, y) ↔ Vector $x\mathbf{i} + y\mathbf{j}$
2. MAGNITUDE:	$ v = \sqrt{x^2 + y^2}$
3. ADDITION:	$(a+c)\mathbf{i} + (b+d)\mathbf{j} = \mathbf{A} + \mathbf{B}$
4. SUBTRACTION:	$(a-c)\mathbf{i} + (b-d)\mathbf{j} = \mathbf{A} - \mathbf{B}$
5. SCALAR MULTIPLICATION:	$k\mathbf{v} = (kx)\mathbf{i} + (ky)\mathbf{j}$
6. PARALLEL VECTORS:	$\mathbf{A} \parallel \mathbf{B}$ jika $\mathbf{A} = k\mathbf{B}$
7. EQUAL VECTORS:	$\mathbf{A} = \mathbf{B}$ jika i-components sama DAN j-components sama

TECHNICAL SPECIFICATIONS

File Format:

- Single HTML file (self-contained)
- No external dependencies (except CDN)
- Works offline once loaded

Responsive Design:

- Mobile (320px): Single column, stacked elements
- Tablet (768px): 2-column where appropriate
- Desktop (1024px+): Full layout with sidebars

Color Scheme:

CSS

```
--primary-blue: #3b82f6 (for i-direction, buttons)
--vector-red: #ef4444 (for j-direction, errors)
--success-green: #22c55e (for correct answers)
--warning-orange: #f59e0b (for hints)
--bg-light: #f9fafb (page background)
--bg-white: #ffffff (cards)
--text-dark: #1f2937 (main text)
--text-muted: #6b7280 (secondary text)
```

Accessibility:

- ✓ WCAG 2.1 Level AA compliant
- ✓ Keyboard navigation (Tab, Enter, Arrows)
- ✓ Color contrast ratio $\geq 4.5:1$
- ✓ ARIA labels on all interactive elements
- ✓ Focus indicators visible (blue outline 3px)
- ✓ Semantic HTML5 structure
- ✓ Screen reader compatible

Performance:

- ✓ File size < 100KB (target: 50KB)
- ✓ Load time < 2 seconds
- ✓ Animations 60fps (no lag)
- ✓ No console errors
- ✓ Zero layout shift

LEARNING OUTCOMES ASSESSMENT

C1 (Remember):

- Students identify scalar vs vector correctly
- Recognize vector notation in multiple forms
- Distinguish vector types (null, unit, parallel, equal)

C2 (Understand):

- Explain why direction matters for vectors
- Express vectors in component form with reasoning
- Interpret geometric representations of vectors
- Explain how vector addition works geometrically

C3 (Apply):

- Solve vector addition/subtraction problems
 - Apply knowledge to real-world scenarios
 - Determine if vectors are parallel or equal
 - Calculate resultant forces/velocities
-

✅ QUALITY CHECKLIST

Before generating, verify this prompt covers:

- ✅ All subtopics from Unit 5 syllabus (5.1, 5.2, 5.3)
 - ✅ C1, C2, C3 competency levels
 - ✅ 2 real-world stories (airplane, building forces)
 - ✅ CVSA progression (Concrete → Visual → Symbol → Abstract)
 - ✅ 4 main interactive patterns fully specified
 - ✅ 6-tab structure with content for each
 - ✅ Difficulty levels (Easy / Medium / Hard)
 - ✅ Hints 4-level progressive system
 - ✅ Full jalan kerja solutions
 - ✅ Dynamic problem generator specifications
 - ✅ Color-coding consistency (blue=i, red=j, green=resultant)
 - ✅ Bilingual (Bahasa Malaysia + English)
 - ✅ Technical requirements (responsive, accessible, performance)
 - ✅ Gamification mechanics (XP, levels, badges, progress)
 - ✅ Common misconceptions addressed with specific feedback
-

🚀 READY TO GENERATE

This prompt is production-ready for Claude code generation.

Next step: Copy this entire prompt (or sections as needed) and paste into Claude chat with command:

```
"Generate complete interactive HTML app for VECTOR topic using this detailed prompt.  
Output: Single .html file, production-ready, all features working."
```

Expected output: Complete, tested, classroom-ready interactive learning application in 5-10 minutes.

End of Detailed VECTOR Prompt